



Introduction to R & R Studio



Part 1. Introduction

Why R....?

- It's free
- Popular worldwide
- It's free
- Lots of easy to access help, tutorials and forums
- It's free
- Most statistical problems can be answered using available packages, and new packages can be created and shared with all users.
- It's free
- Leads to improved repeatability
- It's free

R script

The R script is where you can write and edit your code before running it in the R console. This script can then be saved for future use using "File > Save As".

#Note that lines starting with a hashtag, are not executed in the R console.

#Hashtag allows you to write useful comments explaining your code as required.

R console

Code that you wish to run can be transferred to the console by highlighting the code and selecting “Ctrl+Enter”.

Vectors

Vectors in R can be created using:

```
name_of_vector<-c(data1,data2...)
```

Example: For a statistics module, seven students achieve the following exam marks: 87,45,50,71,88,56,92.

#Creates a vector called “statmod” containing the exam marks.

```
statmod<-c(87,45,50,71,88,56,92)
```

Functions

Functions in R can be applied using

```
function_name(data_function_to_be_applied_to)
```

Example: Using “summary” command to obtain summary statistics for the statistics module exam marks

```
summary(statmod)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
45.00	53.00	71.00	69.86	87.50	92.00

Some other common functions for summarising data include:

```
mean()
```

```
sd()
```

```
median()
```

```
IQR()
```

mad()
min()
max()
range()
fivenum()

Challenge: For a mathematics module, seven students achieve the following exam marks:

90,35,40,52,56,66,76.

Create a vector called 'mathmod' for the mathematics marks.

Helpful Resources

Official R manual:

<https://cran.r-project.org/manuals.html>

Tutorials:

<http://www.tutorialspoint.com/r/>

Questions and answers:

<https://stackoverflow.com/questions/>

Latest R news (and tutorials):

<https://www.r-bloggers.com/>

And there are so many more... simply Google for assistance!*

*other search engines are available.

Part 2 Sourcing appropriate functions and understanding their usage.

Function help files

The help pages can be accessed with “?” inserted prior to the function name you wish to see more information on.

Example: Assuming the same seven students’ exam marks are recorded for the statistics and mathematics modules above, an appropriate test is to be performed.

?t.test

t.test (statmod, mathmod, paired=TRUE)

```
Paired t-test

data:  statistics and mathematics
t = 2.0049, df = 6, p-value = 0.0918
alternative hypothesis: true difference in means is not equal
to 0
95 percent confidence interval:
 -2.330718 23.473575
sample estimates:
mean of the differences
      10.57143
```

Part 3. Installing packages and communicating with software

Installing packages

When you install R, it comes with the default 'stats' package which provides you with the capability to form basic tasks. However, this may not encompass every task you may ever face, particularly in an ever evolving world. R users have created and shared additional packages for doing everything from more sophisticated statistical tests, graphics and maps, to creating Christmas cards. These packages are easily found by searching what you are looking for on your favourite search engine.

The packages are listed at <https://cran.r-project.org/web/packages>.

These packages can be installed using the `install.packages()` command.

Example: Assume that upon review of your analyses you have been advised to perform the Yuen paired samples test to compare the exam marks between the two modules. This test trims a specified percentage of the data from the tails prior to performing the paired samples t-test. This test is not available on the default 'stats' package in R. But can be performed by installing the relevant package called PairedData.

```
install.packages("PairedData")
```

```
library(PairedData)
```

```
?yuen.t.test
```

```
yuen.t.test(statmod,mathmod,paired=TRUE,tr = 0.2)
```

```
Paired Yuen test, trim=0.2

data:  x and y
t = 1.8068, df = 4, p-value = 0.1451
alternative hypothesis: true difference in trimmed means is no
t equal to 0
95 percent confidence interval:
 -6.654587 31.454587
sample estimates:
trimmed mean of x - trimmed mean of y  12.4
```

Communicating with other software

If you wish to save R data frames to a data file:

Firstly, ensure that you have created the relevant folder in the relevant directory.

Secondly, set the working directory in R to the relevant folder which you wish to save to, using the **setwd()** command.

Thirdly, put all the output to be saved into one data frame.

Finally, after you have saved the output it is often advisable to remove the link between R and the working directory using the **dev.off()** command.

Example: writing the exam scores for mathematics and statistics to the "H:" directory in a folder called "R_stuff" in a .csv format with the file name "exam_scores"

```
setwd("H:\\R_stuff")
```

```
output<-data.frame(statmod, mathmod)
```

```
write.csv(output,file="exam_scores.csv")
```

If in a separate R session you wish to read the file back into R:

```
read.csv("H:\\R_stuff\\exam_scores.csv")
```